

Lab 3: Component Modelling & Architectural Pattern Selection

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System: Pharmacy Expiry & Re-order Dispatch Engine (Problem Statement #15)

Architecture Selection

The design uses a Microservices architecture for the Pharmacy Expiry & Re-order Dispatch Engine.

Architectural Choice

Style	Fit for this system	Verdict
Layered	A layered design can separate inventory, expiry monitoring, re-order logic and data access. However, the functions may have different workloads and release cycles, making independent scaling and deployment more difficult.	Rejected
Client-Server	A centralized server can coordinate pharmacy inventory and dispatch operations, but it can become a single point of failure and a scalability bottleneck as inventory checks and re-order processing increase.	Rejected
Microservices	Inventory, expiry monitoring, re-order management and supplier dispatch can be deployed and scaled independently. This supports fault isolation, independent deployment and better scalability for the pharmacy workflow.	Selected

The architecture has five main components: the Inventory Management Service, the Expiry Monitoring Service, the Re-order Management Service, the Supplier/Dispatch Service and the Pharmacy Database. These services can be deployed as independent microservices and communicate through well-defined interfaces.

Two Reasons

First, independent scaling. Expiry monitoring may perform periodic or batch checks across medicine records, while re-order processing and supplier dispatch can experience different workloads. Microservices allow each service to scale according to its own demand.

Second, independent deployment and fault isolation. The services have distinct responsibilities and can be developed and deployed separately. For example, a change to supplier dispatch logic can be released without requiring changes to inventory or expiry monitoring, while a failure in one service can be isolated from the others.

Security Advantage

Each service can have its own access boundary and credentials. Sensitive pharmacy inventory and medicine information can be protected through controlled service-to-service access, authentication and authorization. Secure communication such as HTTPS/TLS can protect data exchanged between services, reducing unauthorized access and limiting the impact of a compromised service.

Performance Benefit

Microservices enable independent scaling and efficient resource utilization. Services performing time-critical work, such as expiry checks and re-order generation, can be allocated additional resources during peak workloads without scaling the entire system. This can improve the responsiveness of inventory updates, expiry alerts and dispatch requests.